

TryHackMe — Intrusion Detection (IDSevasion)

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Room URL: <https://tryhackme.com/room/idsevasion>

Objective:

This report summarizes the activities performed during the TryHackMe 'Intrusion Detection' room. The goal of the room is to introduce intrusion detection systems (IDS), examine detection of common evasion techniques, and understand how security monitoring and detection can be bypassed or adapted. This document includes screenshots, high-level explanations of steps, observations, and conclusions.

Environment & Setup

This room uses TryHackMe AttackBox/Target machines that are provisioned for a limited period. Typical tools used during the exercises include network scanners, packet capture tools, and IDS systems simulators. The exercises are classroom-style: start the provided machine(s), follow tasks, and validate detection behaviour with screenshots and observations.

Note: This report focuses on observations and explanations rather than providing runnable offensive commands.

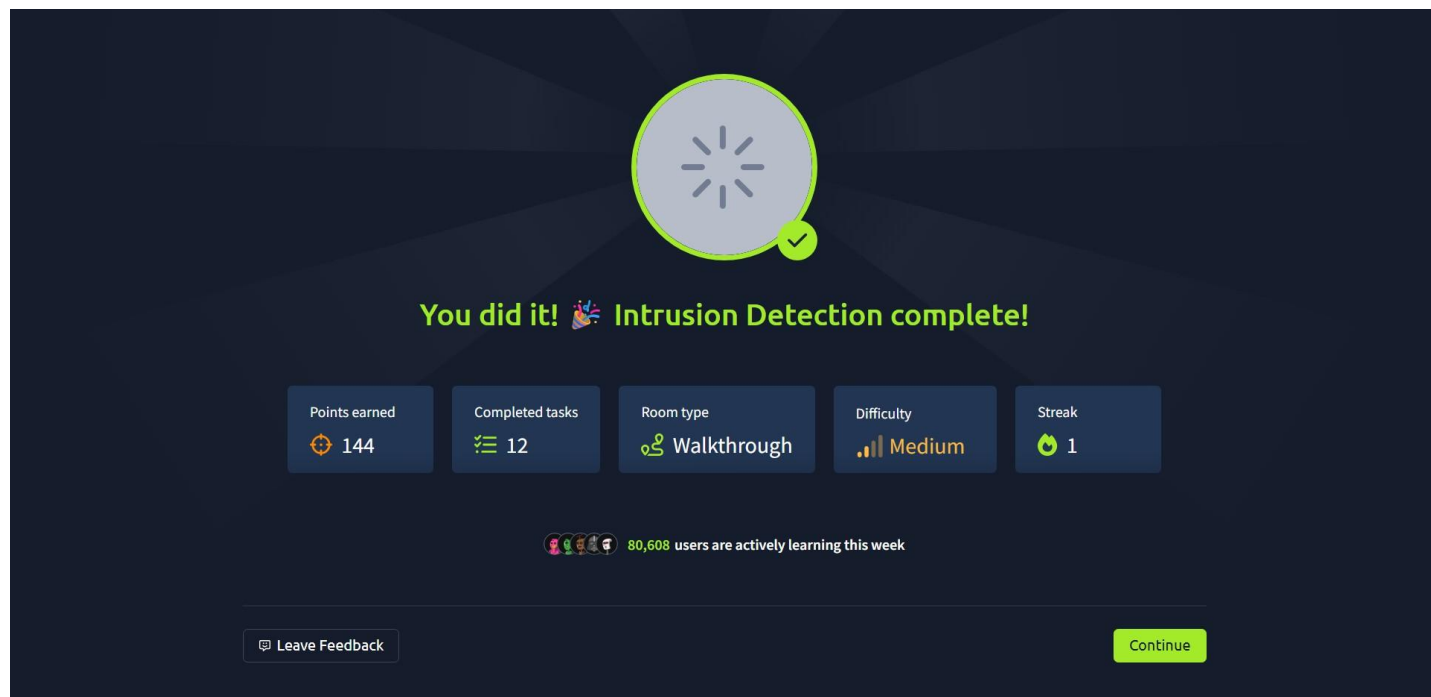


Figure 1: Completion screen for the Intrusion Detection room (points, tasks, difficulty).

Target Machine & Task Overview

The room provides a target machine (an AttackBox/VM) that you start from the room page. The interface shows the target machine title, when the IP will be shown and the remaining time for the machine. Tasks are listed below the target information. The tasks guide you through introduction, detection evasion techniques, and hands-on validation. The screenshot below shows the room page and target machine information.

TryHackMe

Dashboard Learn Practice Compete

Access Machines Go Premium 0

Learn > Intrusion Detection

Intrusion Detection

Learn cyber evasion techniques and put them to the test against two IDS

60 min 10,080

Start AttackBox Save Room 294 Recommend Options

Room progress (0%)

Target Machine Information		
Title	Target IP Address	Expires
DemoCTFFinal	Shown in 0min 17s	59min 17s

Task 1 Introduction

Have you ever completed a CTF and wondered, "Would I have been detected?". This room will serve as an introduction to the world of intrusion detection systems (IDS) and cyber evasion techniques. To complete this room, you will need to orchestrate a full system takeover whilst experimenting with evasion

Start Machine

Figure 2: Room landing page showing target machine information and Start AttackBox button.

High-level Steps Performed (Summary)

1. Read the task instructions on the TryHackMe room page and started the AttackBox/Targetmachine.
2. Performed reconnaissance to identify open services on the target (e.g., scanning and banner observation). This is to determine which services may trigger IDS alerts.
3. Observed IDS behavior: The room simulates IDS rules that flag specific signatures or anomalous activity. You are asked to attempt evasion techniques to see whether the IDS detects the activity.
4. Evasion techniques in the lab (conceptual overview):
 - Fragmentation and packet reassembly evasion (breaking payload across packets).
 - Encoding or obfuscation of payloads so signature-based detection is less likely to match.
 - Timing and throttling to avoid burst patterns that could trigger anomaly detectors.
 - Using legitimate-looking protocols or tunneled channels to hide malicious payloads.
5. Capture and review logs/alerts from the IDS (if provided) to validate whether the attempted activity was detected.
6. Document findings with screenshots and notes (this report).

Observations, Lessons & Conclusion

Observations:

- The TryHackMe room provides a controlled environment to experiment with IDS detection and evasion.
- Signature-based IDS detect known patterns; obfuscation or uncommon fragmentation may reduce detection confidence.
- Modern detection systems often combine signatures with anomaly detection and contextual information, making simple evasion less effective.

Lessons:

- IDS are an important layer of defense but are not foolproof. Defense-in-depth (logs, endpoint security, network controls) is important.
- Evasion exercises should be used for learning and improving detection capabilities, not for malicious activity in the wild.

Conclusion & Recommendations:

- Use the room to learn how different techniques affect detection and to practice generating clear evidence and logs.
- For defenders: enrich IDS with contextual telemetry, use network segmentation, and monitor for subtle indicators.
- Keep signatures and detection rules updated and test them in controlled lab environments regularly.

Attachments and Next Steps

Attached screenshots (embedded earlier) show room completion and machine details. If you upload more screenshots, I can regenerate this PDF to include them inline with captions and specific notes for each task.

Deliverables:

- This PDF report (high-level lab report).
- Optional: Expanded report with per-task screenshots and detailed timeline if you provide additional images.